

Homework 4: Rain and Snow Image Restoration

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<https://github.com/milktea7654/Visual-Recognitionusing-Deep-Learning-2026-Spring>

Abstract. This report describes my solution for the HW4 image restoration competition. The task is to train one model that restores both rain-degraded and snow-degraded images, without external data or pre-trained weights. My final method is a PromptIR-based all-in-one restoration model with a NAF-style encoder-decoder backbone and lightweight Mixed-Scale Feed-Forward Network (MSFN) adapters. The model keeps PromptIR’s degradation-aware prompt idea, but uses feature-conditioned sparse prompt routing inside the decoder. I train the model from scratch with paired random crops, horizontal-flip augmentation, progressive loss scheduling, AdamW optimization, automatic mixed precision, gradient clipping, and exponential moving average weights. The selected checkpoint is from epoch 419 and reaches 31.4245 dB EMA validation PSNR, with 30.6999 dB on rain and 32.1703 dB on snow. Additional experiments show that more complex branches, pseudo masks, and degradation copy-paste were less reliable than the final cleaner PromptIR-NAF-MSFN design, while 2-way horizontal-flip test-time augmentation improved PSNR by about 0.15 dB.

Keywords: Image Restoration · PromptIR · NAFNet · MSFN · Deraining · Desnowing

1 Introduction

The goal of this homework is all-in-one image restoration for two weather degradation types: rain and snow. The training set provides paired degraded and clean images for both degradations. The test set contains only numbered degraded images, so the model cannot rely on test-time labels or separate manually selected rain/snow pipelines. The final system must therefore learn a shared restoration function that can infer the degradation type from the input image.

This task is difficult because rain and snow have different visual structures. Rain streaks are usually thin, directional, and high-frequency, while snow particles are larger, more irregular, and can occlude local image regions. A single model must remove both degradations while preserving background texture. Since the assignment does not allow external data or pretrained weights, every parameter in my model is trained from scratch using only the provided paired rain/snow dataset.

My final approach is named PromptIR-NAF-MSFN. It follows the required PromptIR direction [1] by using learned restoration prompts for all-in-one blind restoration. I modify the restoration backbone into an efficient NAF-style U-Net [2] and add MSFN adapters inspired by DRSformer [3] to improve multi-scale degradation modeling. This modification was motivated by my intermediate experiments: snow PSNR improved more easily, but rain PSNR repeatedly became the bottleneck. Compared with my earlier trials, the final design is intentionally clean: no explicit task classifier, no pseudo-mask head, no contrastive branch, and no degradation copy-paste are kept in the final run.

2 Method

2.1 Data Preprocessing

The dataset contains 1600 rain pairs and 1600 snow pairs for training/validation. Each degraded image has a corresponding clean target. All images are loaded as RGB tensors and normalized to $[0, 1]$. The data loader scans files named `rain-*.png` and `snow-*.png`, then matches them with `rain_clean-*.png` and `snow_clean-*.png`. I use a deterministic split with `val_ratio=0.05`, giving validation examples from both degradation types.

During training, the model receives paired random crops of size 128×128 . If an image is smaller than the crop size, reflection padding is applied before cropping. Horizontal flip is the only geometric augmentation in the final run. Vertical flips and rotations are disabled because rain direction is not rotation-invariant; rotated or upside-down rain creates an unrealistic training distribution.

2.2 PromptIR-Based Architecture

The final model is a single PromptIR-based model for both rain and snow. PromptIR’s key idea is to use learned prompts to encode degradation-aware information, so that one blind restoration network can adapt to multiple degradation types [1]. I keep this idea through decoder-side feature prompt routing rather than an explicit rain/snow classifier.

The model follows a U-Net-like restoration structure. A 3×3 patch embedding maps the input RGB image into feature space. Three encoder levels progressively downsample the features, followed by a latent stage. The decoder upsamples the features and fuses skip connections from the encoder. A refinement stage processes full-resolution features before the output layer. The final RGB prediction is residual: the network predicts a correction and adds it to the degraded input. This is suitable for restoration because many pixels are already close to clean and only degraded regions need large changes.

2.3 Feature-Aware Prompt Routing

Prompt routing is applied at decoder levels 3, 2, and 1. For each decoder feature map, the model computes a global average pooled descriptor and feeds it into a

Component	Final setting
Model name	PromptIR-NAF-MSFN
Input/output channels	3 / 3
Base channel dimension	72
Encoder/latent blocks	[6, 8, 10, 12]
Refinement blocks	8
Prompt length / top- k	5 / 2
Prompt dimensions	[72, 144, 432]
Prompt sizes	[64, 32, 16]
MSFN stages	latent, decoder level 3, decoder level 2, refinement
MSFN blocks per selected stage	1
Approximate parameters	58.72M

Table 1: Final PromptIR-NAF-MSFN model configuration.

lightweight router. The router predicts weights over a prompt bank, but only the top two prompts are kept. The selected prompts are linearly combined, resized to the current feature resolution, projected to the decoder channel dimension, and fused with the decoder feature by convolutional layers.

This feature-conditioned routing allows the same model to adapt its restoration behavior to rain-like and snow-like content without requiring degradation labels at inference time. The final setting uses five prompts, top-2 sparse routing, prompt dimensions [72, 144, 432], prompt sizes [64, 32, 16], and prompt strength 0.10.

2.4 NAF Backbone and MSFN Adapters

The main restoration blocks are NAFNet-style blocks [2]. Each block uses layer normalization, pointwise convolution, depthwise convolution, a simple gate, simplified channel attention, and residual scaling. I chose this family because it is stable and efficient for image restoration from scratch.

To improve multi-scale degradation modeling, I add one MSFN adapter to selected stages: latent, decoder level 3, decoder level 2, and refinement. MSFN is inspired by DRSformer’s mixed-scale feed-forward network for deraining [3]. I introduced this component because the clean PromptIR-NAF direction was stable, but rain PSNR stayed lower than snow PSNR across repeated trials. In my implementation, each adapter processes features through mixed 3×3 and 5×5 depthwise branches before projecting them back to the original channel dimension. The hypothesis is that thin rain streaks, larger snow particles, and background details require different receptive fields, so selected mixed-scale adapters can especially help the rain bottleneck without making every block heavy.

Table 1 summarizes the final model.

Stage	Epochs	Loss	Notes
1	1–50	Charbonnier + 0.04 SSIM + 0.015 FFT	Early structural/frequency stabilization
2	51–260	Charbonnier + 0.01 FFT	Main restoration training
3	261–340	Charbonnier	Augmentation off, LR 3×10^{-5}
4	341–420	MSE	Augmentation off, LR 10^{-5}

Table 2: Progressive training schedule used in the final run.

2.5 Training Strategy

The model is trained from scratch for 420 epochs. I use AdamW, weight decay 10^{-4} , batch size 8, automatic mixed precision, gradient clipping at 1.0, and EMA with decay 0.999. The base learning rate is 1.8×10^{-4} , with 12 warmup epochs from 2×10^{-5} and cosine decay to 10^{-6} when no stage-specific learning rate overrides it.

The final run uses a progressive loss schedule. In the first 50 epochs, the loss is Charbonnier plus small SSIM and FFT amplitude terms. The SSIM term [5] encourages structural consistency, while the FFT term is motivated by frequency-domain reconstruction losses [6]. From epoch 51 to 260, the SSIM term is removed and only a small FFT term remains. From epoch 261 to 340, augmentation is disabled and the model trains with clean Charbonnier loss at learning rate 3×10^{-5} . From epoch 341 to 420, augmentation remains disabled and MSE is used at learning rate 10^{-5} , because the evaluation metric PSNR is directly related to MSE.

2.6 Inference and Submission

The final inference uses the EMA weights from the best checkpoint. Images are restored by tiled inference with tile size 128 and overlap 32. Overlapping tiles are merged using Hann-window weighting, which reduces boundary artifacts. The final test-time augmentation is a two-way gravity-aware TTA: original image plus horizontal flip. This setting matches the horizontal-flip augmentation used during training and improved PSNR by about 0.15 dB compared with using the original image alone. The restored images are clipped to $[0, 1]$, converted to `uint8`, stored in CHW format, and saved in `pred.npz` with keys equal to the original test filenames.

3 Results

The best selected checkpoint is epoch 419. It achieves 31.4245 dB overall EMA validation PSNR, with 30.6999 dB for rain and 32.1703 dB for snow. The final epoch 420 is almost identical at 31.4230 dB, so the model remains stable at the end of training.

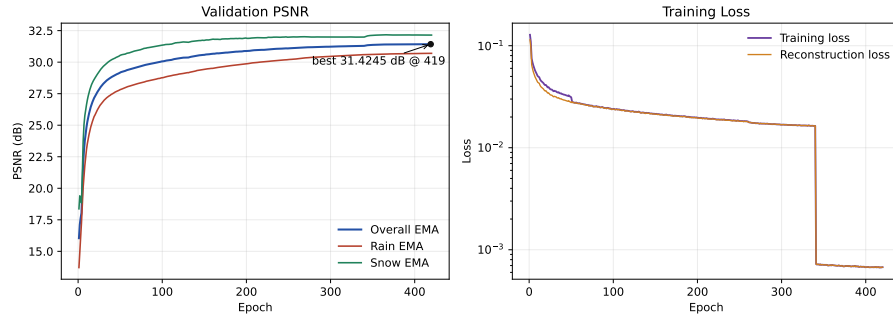


Fig. 1: Final HW4 training curves extracted from the TensorBoard event file. Left: EMA validation PSNR for overall, rain, and snow subsets. Right: training loss and reconstruction loss on a log scale. The best overall validation PSNR is 31.4245 dB at epoch 419.

Metric	Epoch	Value
Best overall EMA PSNR	419	31.4245 dB
Best rain EMA PSNR	419	30.6999 dB
Best snow EMA PSNR	365	32.1703 dB
Final overall EMA PSNR	420	31.4230 dB
Final training loss	420	0.0006745
Lowest training loss	416	0.0006689
Lowest validation Laplacian L1	340	0.03977

Table 3: Summary of final validation and training metrics.

The curves show that validation PSNR improves steadily during the early and middle stages, then becomes very stable after the late fine-tuning phase. Snow restoration is consistently easier than rain restoration. This matches qualitative observation: snow particles are often removed cleanly, while rain images can still lose fine background texture after streak removal. The rain/snow gap suggests that future work should focus on preserving high-frequency background details in rain images.

4 Additional Experiments

The homework requires experiments beyond simple hyperparameter tuning. I tested several architectural, loss, augmentation, and inference ideas before choosing the final model.

PromptIR-CPLFreq baseline. This was an early stable PromptIR-style baseline with frequency-related conditioning. The hypothesis was that frequency information could help restore thin rain streaks and snow texture. It was simple and reliable, and earlier clean runs reached roughly the low 31 dB range. However, it did not have enough capacity for the final attempt, especially on

rain-degraded backgrounds, which still looked blurred. I kept it as a reference but did not use it for final submission.

DFP, contrastive learning, and prototype routing. I tested feature perturbation and degradation-representation routing to reduce interference between rain and snow. The hypothesis was that explicitly separating degradation embeddings would help the shared model learn different restoration behaviors. In practice, batch contrastive learning was unreliable with small batches because there were often not enough positive and negative pairs. Prototype routing reduced part of this issue, but the added complexity did not produce a stable improvement. These modules were rejected.

Degradation copy-paste. I tested internal degradation copy-paste by transferring degradation residuals between paired images. The goal was to increase data diversity without external data. This was risky because rain and snow are not always simple additive zero-mean residuals; some settings produced unnatural dark or bright boundaries. Because this distribution mismatch could hurt PSNR, copy-paste was disabled in the final run.

Pseudo-mask and detail losses. I tested pseudo-mask, phase/detail, and rain-specific losses to preserve clean background regions. These losses were sensitive. They sometimes downweighted severe degraded regions that actually needed restoration, or introduced noisy colors around removed rain/snow. Since the final clean loss schedule was more stable, these auxiliary losses were removed.

DeepMSFN and DetailSAM. I explored MSFN-style mixed-scale modules because rain PSNR was the main bottleneck in my earlier runs. The useful part of this direction was adding lightweight MSFN adapters to the final PromptIR-NAF model. I also designed a DetailSAM probe inspired by MPR-Net’s original-resolution refinement and supervised-attention motivation [7]. The protocol initialized from an existing PromptIR-DeepMSFN checkpoint, verified zero-initialized equivalence to the baseline, froze the backbone for a short period, and then fine-tuned jointly. The hypothesis was that an original-resolution detail branch could recover water ripples, fur, building edges, and leaves without disturbing the main restoration mapping. The DetailSAM branch was useful as a design study, but it did not clearly beat the cleaner PromptIR-NAF-MSFN direction, so it was not selected.

TTA ablation. I compared inference without TTA and inference with horizontal-flip TTA. The 2-way setting improved PSNR by about 0.15 dB, so I used it in the final submission. I also tried 8-way TTA late in development, but this was not a fair match for the trained model because the final training pipeline did not include rotations or vertical flips. As a result, rotated rain patterns became out-of-distribution. The conclusion is not that 8-way TTA is generally bad, but that inference-time geometry should match the augmentation distribution used during training.

Variant	Finding	Final decision
PromptIR-CPLFreq	Stable but limited capacity	Reference only
DFP / contrastive / prototype	Unstable with small batches	Rejected
Degradation copy-paste	Distribution mismatch risk	Disabled
Pseudo-mask/detail losses	Sensitive and artifact-prone	Disabled
DetailSAM	No stable PSNR gain	Rejected
PromptIR-NAF	Most reliable intermediate direction	Basis
PromptIR-NAF + MSFN	Addressed rain PSNR bottleneck	Final model
2-way hflip TTA	About +0.15 dB PSNR	Used in final inference

Table 4: Summary of additional experiments and final decisions.

5 Conclusion

This project implements a single PromptIR-based model for all-in-one rain and snow restoration. The final PromptIR-NAF-MSFN design combines feature-conditioned prompt routing, efficient NAF-style restoration blocks, and selected MSFN adapters for multi-scale degradation modeling. The final model is trained from scratch using only the provided data and reaches 31.4245 dB overall EMA validation PSNR. The main lesson from the experiments is that a clean and stable PromptIR-based restoration model outperformed several more complicated auxiliary branches. The main remaining weakness is rain restoration, where preserving background texture after streak removal is still harder than snow removal.

References

1. Potlapalli, V., Zamir, S.W., Khan, S.H., Khan, F.S.: PromptIR: Prompting for all-in-one image restoration. In: NeurIPS (2023)
2. Chen, L., Chu, X., Zhang, X., Sun, J.: Simple baselines for image restoration. In: ECCV, pp. 17–33 (2022)
3. Chen, X., Li, H., Li, M., Pan, J.: Learning a sparse transformer network for effective image deraining. In: CVPR, pp. 5896–5905 (2023)
4. Zamir, S.W., Arora, A., Khan, S., Hayat, M., Khan, F.S., Yang, M.H.: Restormer: Efficient transformer for high-resolution image restoration. In: CVPR, pp. 5728–5739 (2022)
5. Wang, Z., Bovik, A.C., Sheikh, H.R., Simoncelli, E.P.: Image quality assessment: From error visibility to structural similarity. IEEE Transactions on Image Processing 13(4), 600–612 (2004)
6. Jiang, L., Dai, B., Wu, W., Loy, C.C.: Focal frequency loss for image reconstruction and synthesis. In: ICCV, pp. 13919–13929 (2021)
7. Zamir, S.W., Arora, A., Khan, S., Hayat, M., Khan, F.S., Yang, M.H., Shao, L.: Multi-stage progressive image restoration. In: CVPR, pp. 14821–14831 (2021)
8. Yoo, J., Ahn, N., Sohn, K.A.: Rethinking data augmentation for image super-resolution: A comprehensive analysis and a new strategy. In: CVPR, pp. 8375–8384 (2020)

9. Zhang, H., Cisse, M., Dauphin, Y.N., Lopez-Paz, D.: mixup: Beyond empirical risk minimization. In: ICLR (2018)
10. Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., Krishnan, D.: Supervised contrastive learning. In: NeurIPS (2020)